

AIRCRAFT MAINTENANCE MANUAL

& ILLUSTRATED PARTS CATALOG (IPC)

BOEING 747-8F FLEET

DOCUMENT NUMBER: B748-AMM-IPC-001

VOLUME 1: SYSTEMS AND ALTERNATE CONFIGURATIONS

WARNING: THIS DOCUMENT CONTAINS SYNTHETIC TEST DATA
AUTHORIZED FOR HACKATHON AND PROTOTYPING USE ONLY.
DO NOT USE FOR ACTUAL AIRCRAFT MAINTENANCE.

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1 CHAPTER 01 - STANDARD PRACTICES & SAFETY

1.1 01-10-00: General Safety Warnings

Maintenance personnel must adhere to all safety warnings outlined in this manual. Failure to comply may result in injury, death, or damage to the airframe.

WARNING: BEFORE STARTING MAINTENANCE, MAKE SURE THAT THE AIRCRAFT IS SAFELY GROUNDED.

CAUTION: DO NOT SUPPLY ELECTRICAL POWER TO THE AIRCRAFT UNLESS ALL FLIGHT CONTROL SURFACES ARE CLEAR OF PERSONNEL AND EQUIPMENT.

1.2 01-20-00: Alternate Parts Policy (General)

This manual details the primary and approved alternate components for the 747-8F airframe. When a primary part is unavailable, mechanics must consult the specific chapter for allowable substitutions.

Classifications of alternate parts include:

- **True Alternate (Direct Fit):** Requires no structural changes.
- **Oversized Version:** Requires specific brackets, shims, or hardware kits due to airframe wear.
- **Optional Fit:** Conditionally approved based on operational parameters (e.g., restricted MTOW, ferry flights).

No deviations from these classifications are permitted without prior Engineering Liaison (EL) team approval.

2 CHAPTER 32 - LANDING GEAR

2.1 SUB-SECTION 32-30-01: MAIN LANDING GEAR RETRACTION ACTUATOR

2.1.1 General Description

This section provides the approved configuration data and alternate part specifications for the Main Landing Gear (MLG) Retraction Actuator Assembly.

2.1.2 Primary Part Specification

- **Nomenclature:** ACTUATOR ASSY - MLG RETRACT
- **Primary Part Number:** ACT-747-PRM
- **Supplier:** Boeing Component Operations
- **Status:** Active

2.1.3 Approved Alternate Parts (Configuration & Fitment)

In the event that the primary part (ACT-747-PRM) is unavailable in AMAP or Maintenix stock, the following alternate components are safely approved for installation on the 747-8F fleet:

A. True Alternate (Direct Fit)

- **Part Number:** ACT-747-ALT1
- **Classification:** True Alternate
- **Technical Notes:** This part is a direct bolt-on replacement and is 100% cross-compatible. No structural modifications or additional engineering approvals are required. Fully certified for standard maximum takeoff weight (MTOW).

B. Oversized Version

- **Part Number:** ACT-747-OVR
- **Classification:** Oversized Version
- **Technical Notes:** This is an oversized actuator designed for airframes experiencing trunion wear. Installation requires the addition of structural modification bracket B-99 and the use of oversized titanium bolts (Hardware Kit HK-22). Maintenance teams must refer to Structural Repair Manual (SRM) 32-00-05 before installation.

C. Optional Fit

- **Part Number:** ACT-747-OPT

- **Classification:** Optional Fit
- **Technical Notes:** This is an optional fit actuator originally designed for the older 747-400 ERF fleet. It is safe and approved for the 747-8F ONLY for emergency ferry flights or lightweight cargo operations (MTOW restricted to 850,000 lbs). It requires an Engineering Liaison (EL) team sign-off prior to flight dispatch.

3 CHAPTER 21 - AIR CONDITIONING

3.1 SUB-SECTION 21-50-01: AIR CYCLE MACHINE PACK VALVE

3.1.1 General Description

This section outlines the flow control pack valves responsible for regulating bleed air into the Air Cycle Machine (ACM).

3.1.2 Primary Part Specification

- **Nomenclature:** VALVE ASSY - PACK FLOW CONTROL
- **Primary Part Number:** VLV-ACM-990
- **Status:** Active

3.1.3 Approved Alternate Parts

A. True Alternate (Direct Fit)

- **Part Number:** VLV-ACM-991-ALT
- **Classification:** True Alternate
- **Technical Notes:** Direct fit replacement. Software update to the Pack Controller (PC) is recommended but not mandatory for dispatch.

B. Optional Fit (High Altitude)

- **Part Number:** VLV-ACM-HA
- **Classification:** Optional Fit
- **Technical Notes:** High-altitude specialized valve. Approved for use only when operating out of airports with field elevations exceeding 8,000 ft MSL.

4 CHAPTER 27 - FLIGHT CONTROLS

4.1 SUB-SECTION 27-11-05: INBOARD AILERON POWER CONTROL UNIT

4.1.1 General Description

The Inboard Aileron Power Control Unit (PCU) provides hydraulic actuation for the primary roll control surfaces during low-speed and high-speed flight regimes.

4.1.2 Primary Part Specification

- **Nomenclature:** PCU ASSY - INBD AILERON
- **Primary Part Number:** PCU-AIL-INB-01
- **System:** Hydraulic Systems 1, 2, and 3

4.1.3 Approved Alternate Parts

A. True Alternate (Direct Fit)

- **Part Number:** PCU-AIL-INB-02A
- **Classification:** True Alternate
- **Technical Notes:** Manufactured by a secondary vendor. Direct fit. Ensure hydraulic lines are bled completely after installation in accordance with AMM 27-11-00.

B. Oversized Version

- **Part Number:** PCU-AIL-INB-03-OS
- **Classification:** Oversized Version
- **Technical Notes:** Oversized rod-end bearing. Requires re-reaming of the aileron attach lug. Only permitted if lug wear exceeds 0.050 inches.

5 CHAPTER 28 - FUEL SYSTEMS

5.1 SUB-SECTION 28-22-10: CENTER WING TANK BOOST PUMP

5.1.1 General Description

The Center Wing Tank (CWT) Override/Jettison boost pumps supply pressurized fuel to the engine manifold and are capable of jettisoning fuel in flight.

5.1.2 Primary Part Specification

- **Nomenclature:** PUMP ASSY - CWT OVERRIDE
- **Primary Part Number:** PMP-CWT-88A
- **Electrical:** 115V AC, 400Hz, 3-Phase

5.1.3 Approved Alternate Parts

A. True Alternate (Direct Fit)

- **Part Number:** PMP-CWT-88B
- **Classification:** True Alternate
- **Technical Notes:** Identical form, fit, and function. This part features an upgraded impeller shaft seal. No operational restrictions apply.

WARNING: FUEL TANK ENTRY REQUIRES PURGING AND VENTILATION PROCEDURES AS OUTLINED IN AMM 28-10-00. CONFINED SPACE PROTOCOLS MUST BE FOLLOWED.

6 CHAPTER 49 - AIRBORNE AUXILIARY POWER

6.1 SUB-SECTION 49-10-22: APU STARTER GENERATOR

6.1.1 General Description

The Auxiliary Power Unit (APU) Starter Generator is utilized to spool the APU turbine during starting sequences and provides AC power to the electrical buses when the APU is running at governed speed.

6.1.2 Primary Part Specification

- **Nomenclature:** STARTER/GEN ASSY - APU
- **Primary Part Number:** SG-APU-90kVA
- **Output:** 90 kVA

6.1.3 Approved Alternate Parts

A. Optional Fit

- **Part Number:** SG-APU-120kVA-ALT
- **Classification:** Optional Fit
- **Technical Notes:** This is a higher-capacity 120 kVA generator. It is physically larger. To install, the APU aft cowl support strut (Part Number: ST-992) must be temporarily removed. Once installed, it is fully approved for unlimited operations.

7 CHAPTER 73 - ENGINE FUEL AND CONTROL

7.1 SUB-SECTION 73-11-05: MAIN ENGINE FUEL PUMP (GENx-2B)

7.1.1 General Description

This section applies to the GENx-2B engines fitted to the 747-8F. The Main Engine Fuel Pump (MFP) is a dual-stage, engine-driven pump that supplies high-pressure fuel to the Fuel Metering Unit (FMU).

7.1.2 Primary Part Specification

- **Nomenclature:** PUMP ASSY - MAIN ENGINE FUEL
- **Primary Part Number:** PMP-GENX-MFP1
- **Status:** Active

7.1.3 Approved Alternate Parts

A. True Alternate (Direct Fit)

- **Part Number:** PMP-GENX-MFP2
- **Classification:** True Alternate
- **Technical Notes:** Fully interchangeable with MFP1. Note that the spline drive must be lubricated with Engine Oil (MIL-PRF-23699) prior to mating with the accessory gearbox.

CAUTION: DO NOT ROTATE THE PUMP DRY. ENSURE FUEL IS PRESENT IN THE INLET LINE BEFORE MOTORING THE ENGINE.

8 APPENDIX A: GLOSSARY AND TORQUE LIMITS

8.1 Standard Torque Values

Unless otherwise specified in the specific AMM chapter, use the following standard torque values for hardware installations:

- 1/4-28 UNF: 50-70 in-lbs
- 5/16-24 UNF: 100-140 in-lbs
- 3/8-24 UNF: 160-190 in-lbs
- 1/2-20 UNF: 400-480 in-lbs

8.2 Acronym Glossary

- **ACM:** Air Cycle Machine
- **AMM:** Aircraft Maintenance Manual
- **APU:** Auxiliary Power Unit
- **CWT:** Center Wing Tank
- **EL:** Engineering Liaison
- **FMU:** Fuel Metering Unit
- **IPC:** Illustrated Parts Catalog
- **MLG:** Main Landing Gear
- **MTOW:** Maximum Takeoff Weight
- **PCU:** Power Control Unit
- **SRM:** Structural Repair Manual

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